

# *Vi begåfvo oss på väg:* Exploring *jag* and *vi* in Historical Swedish Migration Discourse

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## Abstract

This study investigates first-person pronouns' encoding of social positioning in nineteenth-century letters written by Swedish migrants in Brazil. Drawing on theories of pronouns as markers of social identity, the analysis looks for patterns in the use of *jag* (I) and *vi* (we) across 44 letters, as well as differences in syntax and semantics triggered by the two pronouns. The methodology integrates quantitative measures of pronoun frequency with qualitative analyses of sentence-level semantics (topic modeling, lexical patterns) and syntax (part-of-speech distributions). The results partly align with De Fina (2003) and reveal a diachronic shift from collective (*vi*) to individual (*jag*) self-positioning between two major migration waves. Semantic and syntactic contrasts suggest that *jag*- and *vi*-sentences are structured differently and that the former tend to foreground personal perception and reflections; whereas the latter more often describe action and the surrounding environment.

## 1 Introduction

The analysis of pronoun usage has long been a topic of interest in linguistics, due to the fact that pronouns are the language unit that best encodes the connection between the storytelling-world and the real world. Early research includes Benveniste (1971), who introduced pronouns as means to connect linguistic expression to a specific context. More recent studies claim that pronouns deliver the "social identities of participants or the social relationships between them" (Levinson 1983, p. 89), or in other words, they "provide[s] a window to the relationship between selves and the outside world" (Mühlhäusler and Harré 1990, p. 207). They thus act as markers of social positioning, or of how speakers position themselves and the referents of their discourse in relation to broader social contexts. On this basis, pronouns have also been studied by anthropologists and psychologists, who linked

them to the concept of self-determination of individuals in different cultures (Matsumoto 1994; Duranti 1993).

Drawing on this theory, the present study looks at nineteenth century letters of Swedish migrants in Brazil and their use of first person pronouns, *jag* and *vi*, as they represent the two ends of the social continuum between individualism and collectivism. The first aim of this paper is to reveal general social stance patterns, i.e. to detect preference of one pronoun over the other. In addition, the paper aims at determining the syntactic and semantic differences between *jag*- and *vi*-sentences, that is to say the linguistic difference in the expression of social positioning. The research questions can thus be defined as:

- RQ1: Is there a visible pattern in the usage of *jag* and *vi* across time?
- RQ2: What are the syntactic and semantic differences that characterize *jag*- versus *vi*-sentences?

## 2 Background

It is safe to say that the use of personal pronouns in discourse has been extensively pursued from a wide range of theoretical and methodological perspectives across several disciplines, including linguistics, sociology, psychology and anthropology. As mentioned in the introductory section, authors like Benveniste (1971), Levinson (1983), Mühlhäusler and Harré (1990) but also Brown and Gilman (1972), Urban (1989) and many others repeatedly showed how pronouns are functions that map relationships among participants in a discourse.

The present study is based on the textual analysis carried out by De Fina (2003, p. 51), who draws on these theories to determine how "narrators present themselves in relation to others in stories of personal experience". Letters thus suit this study best,

since they are perfect examples of narration of personal experience. By studying the use of pronouns in stories told by Mexican immigrants in the U.S., De Fina (2003) investigates the authors' social positioning. She finds that "Mexican immigrants display in their narrative discourse an orientation to others that is exemplified by linguistic phenomena connected with the use of pronouns" (p. 89), and that they show a collectively oriented social identity. Similarly, this study hypothesizes the existence of patterns of social positioning in the letters' narration and attempts to reveal them.

Based on the further hypothesis that *jag*- and *vi*-sentences differ syntactically and semantically, this paper also analyses the contrasts between the encoding of social positioning - individual or collective authorial stance - at the sentence-level. While the intent is reminiscent of De Fina (2003)'s work, the techniques used differ from the ones presented in her paper. They combine quantitative and qualitative methods in line with recent text-analytic work that uses Natural Language Processing (NLP) and Machine Learning (ML), as for instance does the study conducted by Devinney, J. Björklund, and H. Björklund (2020), who analyse personal pronouns and gender-related lexical items to detect gender bias. Their use of topic modeling - a technique to uncover semantic patterns in documents also employed here - in both a quantitative and qualitative analysis offers a useful methodological foundation for the current investigation.

Wang, Tseng, and Johanson (2021) similarly explore the use of the pronoun "we" in abstracts and conclusions of scientific papers, integrating qualitative observations with quantitative measures such as frequency counts, log-likelihood values, concordance, pattern analysis, and clustering; enabling a multifaceted perspective on the data.

Other studies have applied related approaches to different linguistic phenomena. Säily, Mäkelä, and Hämäläinen (2021) trace the emergence of neologisms in historical English correspondence using a hybrid automated-manual approach, while Brodén et al. (2025) investigate compounds and neologisms in political discourse through quantitative frequency analysis followed by qualitative semantic grouping. Both show how such methods can uncover new insights into historical change.

While the methodological techniques employed in this study are therefore not novel, their application to historical Swedish correspondence represents an original contribution. To the best of my

knowledge, no prior research has undertaken this type of analysis on this material. The study therefore has the potential to reveal new linguistic and cultural insights into these documents.

### 3 Data and method

The data comprises a set of letters written by Swedish migrants in São Paulo, Brazil, between 1869 and 1892. They were collected from the [Swedish National Library](#)'s database by professor L. Lopez of Stockholm University. These letters were published in Swedish newspapers and were originally written as 44 distinct pieces, two of which were divided and released across consecutive issues, resulting in a corpus of 46 texts. The letters correspond to two migration waves that occurred before and after 1880. This division is reflected in the corpus, with 10 texts originating from the first wave and 36 from the second. All manipulation of data was done using custom Python code, unless otherwise stated. Graphs and plots (not tables) were created using the library Matplotlib (Hunter 2007).

The methodology in this study combines quantitative and qualitative analysis, broadly aligning with the works discussed in Section 2. The quantitative component focuses on pronoun frequencies and related statistics, whereas the qualitative analysis encompasses topic modeling, the examination of nominal and verbal lemmas and the analysis of part-of-speech (PoS) tags. Prior to either analysis, however, preprocessing was required. For each letter only the strictly narrative text was manually extracted, and any material added by the newspaper, as well as the narrator's signature, was removed. This was done to ensure a cleaner dataset. All omitted lines were nonetheless documented in a separate file to maintain replicability. Superfluous newlines and spaces were removed due to tokenization issues, and filenames were converted into simpler identifiers to facilitate the presentation of results.

#### 3.1 Quantitative analysis

As mentioned above, the quantitative analysis centers on pronoun frequency. Tokenization and PoS-tagging were performed using the publicly available Stanza (Qi et al. 2020) library. The Swedish Stanza models used all score above 97 in their [accuracy metrics](#). Although they are trained on modern rather than nineteenth-century Swedish, their performance proved sufficient for this step: man-

ual inspection of the tokenization confirmed strong performance, and PoS-tagging - while occasionally imperfect - was adequate for identifying *jag* and *vi*, whose forms are unchanged across periods. The analysis begun with the calculation of global metrics such as the total number of sentences as well as the count and percentage of *jag*- and *vi*-sentences per letter. The results were ordered chronologically and visualized in a bar chart (Figure 3). This allowed differences in both pronoun usage and overall sentence volume to be considered. Percentage values were used to assess authors' pronoun preferences, in line with De Fina (2003).

For the last step in the quantitative analysis, the total occurrences of *jag* and *vi* across all letters for each migration wave were calculated alongside the overall word count. The frequencies per 1000 words of both pronouns were then computed and compared.

### 3.2 Qualitative analysis

This step figures a multifaceted analysis of *jag*- and *vi*-sentences. The first component is topic modeling, which required training an embedding model to obtain a numerical representation of the sentences. Publicly available models could not be used, as they are predominantly trained on modern Swedish. For the sake of simplicity and methodological transparency a fastText (Bojanowski et al. 2016) model was trained from scratch for 50 epochs. This particular model was chosen over alternatives such as Word2Vec (Mikolov et al. 2013) because its character-based architecture is better suited to historical Swedish, whose orthography often diverges not only from modern Swedish but also varies from author to author. The training corpus consisted of all letters used in this study, supplemented by an 1896 diary, also provided by Professor L. Lopez of Stockholm University. As with the letters, the diary was cleaned by removing non-narrative elements - which were collected in a separate file for replicability. Even after combining these materials, however, the corpus contained fewer than 3000 sentences, making it a very small dataset for training an embedding model. This limitation likely resulted in relatively diffuse embeddings. To reduce noise, punctuation was removed and all tokens were lowercased.

All *jag*- and *vi*-sentences were then extracted from the letters, stripped of punctuation, and processed by the model to generate embeddings. These were then clustered into *jag* and *vi* semantic

maps using PCA (Principal Component Analysis) and KMeans clustering from the library scikit-learn (Pedregosa et al. 2018) - which reduce embeddings to two-dimensional points and then cluster them in K groups. To ensure replicability, a fixed seed of 42 was employed during PCA and KMeans clustering, as well as when training the fastText (Bojanowski et al. 2016) model. Several iterations were needed to identify stopwords, whose removal is essential to generate embeddings with reduced noise and more focused semantic content. Publicly available stopword lists could not be used, as they are based on modern Swedish. After finalizing the list and applying it to the sentences input to the embedding model, the program was run again to generate separate semantic maps of *jag*- and *vi*-sentences for K = (1,2,3). For each value of K, the ten most frequent words in the cluster were extracted to inspect the semantic focus.

A semantic-syntactic analysis followed. The first step involved examining nominal and verbal patterns: *jag*- and *vi*-sentences were tokenized and lemmatized using Stanza (Qi et al. 2020) models, and lists of the most frequent - i.e. occurring more than ten times - nouns and verbs were extracted for both categories. Lemmatization was chosen because the analysis aimed to capture underlying meanings rather than surface-level word forms. However, the model's modern-Swedish training led to errors with historical spellings, particularly for verbs. To address this issue, a custom implementation of the Levenshtein algorithm (see Jurafsky and Martin 2025, section 2.8) was employed to identify the closest matches for each top-ten verb and manually correct the list. High-frequency functional verbs such as "ha" and "få" were excluded from this analysis due to their limited semantic contribution.

Finally, a part-of-speech analysis was conducted using another Stanza (Qi et al. 2020) model to reveal which PoS-tags most commonly co-occurred with *jag* and *vi*. For this purpose, the sentences were processed to include special <START> and <END> tokens. A five-word window, centered on the pronoun, was extracted from each sentence, and a frequency distribution of PoS-tags was computed separately for each pronoun and window position.

## 4 Results

### 4.1 Quantitative analysis

As noted above, this stage began with the calculation of global metrics. The corpus contains a total

|                            | <i>jag</i> /1000 | <i>vi</i> /1000 | word count |
|----------------------------|------------------|-----------------|------------|
| First wave<br>(1869-1880)  | 3.46             | 10.83           | 11545      |
| Second wave<br>(1880-1892) | 7.92             | 6.62            | 37630      |

Table 1: Frequency distribution of *jag* and *vi* per migration wave.

of 2322 sentences across all letters, of which 265 include *jag* and 301 include *vi*. Their distribution - in chronological order - can be observed in Figure 3, which presents the number of *jag*- and *vi*-sentences in relation to the total number of sentences in each letter.

Regarding pronoun usage, 44 of the 46 texts exhibit a clear predominance of one pronoun over the other - meaning that in most letters, one type of sentence accounts for more than 50% of all analysed sentences. The frequencies of *jag* and *vi* per 1000 words in each migration wave are presented in Table 1, alongside the total word counts per wave.

## 4.2 Qualitative analysis

The topic modeling analysis revealed a pattern in both sentences, shown in Figure 1 and 2. Amongst all K tested, K = 1 turned out to be the best, since for K = (2,3) there was no meaningful clustering.

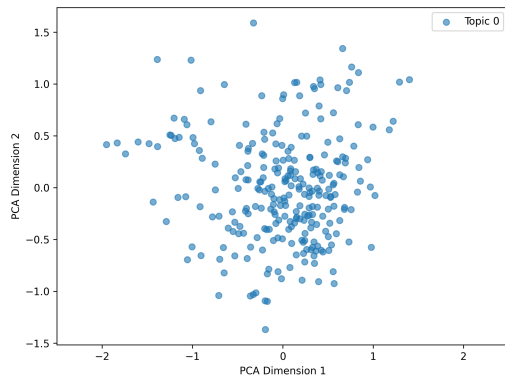


Figure 1: Topic modeling with K=1 for *jag*-sentences.

The ten words most representative of the clusters are as follows: [brasilien, bref, milreis, gång, arbete, svensk, säga, sett, rio, år] for *jag* and [brasilien, dagar, ligga, land, rio, staden, fått, gå, se, ombord] for *vi*. It is important to note that these items depend on the embedding model, which - given the limited amount of training data - is necessarily imprecise and therefore somewhat diffuse.

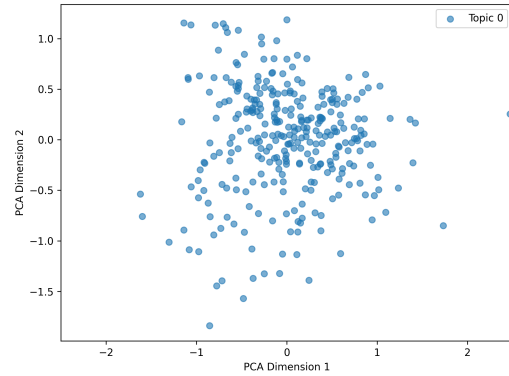


Figure 2: Topic modeling with K=1 for *vi*-sentences.

Consequently, they should not be interpreted as the most frequent lexical items in the corpus, but rather as a reflection of the most common embeddings produced by the model developed in Section 3.2 .

Regarding the analysis of the nominal and verbal domains, the results are presented in Tables 2 and 3. These tables list the most frequent (lemmatized) nouns and verbs occurring in *jag*- and *vi*-sentences, along with their corresponding frequency counts. It is worth noting that four of the verb counts in *jag*-sentences and five in *vi*-sentences required manual adjustment based on the output of the Levenshtein algorithm, in order to correct cases where the lemmatizer failed to group all verb forms under the same lemma.

The n-gram analysis, with n = 5, yielded the results shown in Table 4, normalized as percentages on the total count of PoS-tags (338 for *jag* and 374 for *vi*). An explanation of the tags is available [here](#).

## 5 Discussion

The study revealed both hypotheses in section 2 to be correct. Both analytical components yielded valuable results.

### 5.1 Quantitative analysis

The observation that the vast majority of the narratives (44 out of 46) display a predominance of one pronoun over the other supports the deduction that these authors express a clear position along the spectrum of social stance, in line with the conclusions reached by De Fina (2003). Furthermore, pronoun prevalence seems to shift with the passing of time. As can be noticed in Figure 3, *vi* - collectively oriented self-positioning - is the prevalent pronoun in the early stages of the migra-



tion, whereas *jag* - individualistic social stance - gradually gains prominence over the years and surpasses *vi* toward the end of the second migration wave, even if the totality of *vi*-sentences outnumber *jag*-sentences by 36 occurrences. From a sociolinguistic perspective, this pattern may suggest that Swedish immigrants in Brazil initially depended on community to navigate and adapt to the new environment, but progressively transitioned toward a more individualistic lifestyle as they established themselves in their new social context. This assumption is supported by the findings in Table 1, according to which the frequency of *vi* per 1000 words plummets during the second migration wave. In fact, although the second wave contains more than three times the total word count of the first, its *vi*-rate per 1000 words amounts to only about 60% of the rate observed in the first wave. The quantitative section of the analysis thus answers positively to the first research question: whether there is a visible pattern in the usage of *jag* and *vi* across time.

## 5.2 Qualitative analysis

The qualitative analysis addresses the second research question, i.e. the syntactic and semantic differences that characterize *jag*- and *vi*-sentences. Figures 1 and 2 illustrate these two sentence types in two-dimensional semantic maps as a result of topic modeling, revealing subtle differences in their principal areas of semantic concentration. Even though - as was pointed out in section 3.2 - the embeddings were trained on a corpus too small to ensure high precision, the top ten words extracted from the clusters provide meaningful insight into the semantic orientations of the two sentence types. While both engage with themes related to Brazil ("brasilien", "rio"), *jag*-sentences additionally tend to emphasize commerce ("milreis", a historical brazilian currency) as well as recollections of Sweden and the act of writing letters ("bref", "svensk"). *Vi*-sentences frequently describe the new country and urban environment ("land", "staden") and highlight journey ("dagar", "gå", "se", "ombord").

The frequency analysis of nominal and verbal lemmas can be seen as complementary to the topic modeling results. As shown in Table 2, narration in *jag*-sentences appears to rely heavily on temporal reference ("dag", "gång", "år", "tid"). It also revolves around environment ("land", "hus", "stad", "arbete"), story-telling ("bref") and journey ("milreis", "emigrant"). Similarly, *vi*-sentences ex-

hibit semantic convergence around temporal narration ("dag", "gång", "tid") and references to urban environment ("stad", "land"). Curiously enough, travel-related vocabulary does not appear. Table 3 shows that *jag* is mostly surrounded by verbs related to story-telling ("säga", "skriva"), journey ("komma"), perception ("se", "höra") and cognition ("tro", "veta"). *Vi*, instead, is primarily associated with perception ("se"), then journey ("komma", "gå"), spatial localisation ("ligga") and narration ("säga").

The final considerations derive from the PoS-tags analysis depicted in Table 4. The two-word context surrounding *jag* and *vi* reveals some distinct patterns. *Vi* is most commonly preceded by adverbs, punctuation, verbs and subordinating conjunctions; and followed by verbs, adverbs, adpositions and nouns. By contrast, it is mostly <START> tokens, punctuation, verbs and auxiliaries that precede *jag*; and verb, auxiliaries and adverbs that follow it. These patterns, along with the rest of the findings from the qualitative analysis, are evidence of different narrative functions. *Jag* occurs more easily at the beginning of a sentence and is typically surrounded by narrative, perceptive and cognitive verbal material, aligning it with openings and subjective commentary. *Vi*, which more often co-occurs with subordinating conjunctions, nouns, adverbs and adpositions, appears more integrated into the ongoing flow of narration and event progression.

## 6 Conclusion

This study demonstrated that first-person pronouns can indeed encode authorial social stance and showed how - for these letters - they trace a meaningful temporal pattern, where collective self-positioning gradually fades into a more individualistic view. It was also demonstrated that the two pronouns function differently both semantically and syntactically and participate in distinct narrative structures, with *jag* aligning with perception and subjective story-telling and *vi* encoding progressive narration and descriptions.

## 7 Limitations

It is important to note that the 44 letters studied here are but a small portion of a much larger collection that was unfortunately not available for this study. This analysis should therefore be applied to the entire corpus before any findings can be re-

garded as conclusive. The small number of texts also presented a limitation as training data for the embedding model, as previously discussed. In addition, the results were repeatedly affected by the reliance on language models trained on modern rather than historical Swedish. Further research could therefore focus on improving these limitations.

## 8 Ethical considerations

The data used in this study is public and available on the [Swedish National Library](#)'s website. The files and scripts will be made available upon request. As mentioned in section 3.2, a fixed seed of 42 was employed for replicability and all removed text was saved separately.

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## Appendix

| <i>jag</i>  |              | <i>vi</i>   |              |
|-------------|--------------|-------------|--------------|
| <i>Noun</i> | <i>Freq.</i> | <i>Noun</i> | <i>Freq.</i> |
| dag         | 23           | dag         | 26           |
| land        | 21           | stad        | 24           |
| bref        | 20           | land        | 19           |
| gång        | 20           | gång        | 17           |
| år          | 17           | tid         | 15           |
| milreis     | 16           |             |              |
| arbete      | 15           |             |              |
| tid         | 13           |             |              |
| emigrant    | 12           |             |              |
| hus         | 11           |             |              |
| stad        | 11           |             |              |

Table 2: Nominal frequency analysis in *jag*- and *vi*-sentences.

| <i>jag</i>  |              | <i>vi</i>   |              |
|-------------|--------------|-------------|--------------|
| <i>Verb</i> | <i>Freq.</i> | <i>Verb</i> | <i>Freq.</i> |
| ha          | 42           | ha          | 52           |
| säga        | 28           | få          | 38           |
| se          | 26           | se          | 19           |
| få          | 25           | komma       | 19           |
| tro         | 16           | gå          | 17           |
| höra        | 13           | göra        | 16           |
| göra        | 13           | ligga       | 15           |
| komma       | 13           | säga        | 11           |
| skriva      | 13           |             |              |
| vara        | 13           |             |              |
| veta        | 11           |             |              |

Table 3: Verbal frequency analysis in *jag*- and *vi*-sentences.

| Tag     | TAG1  |       | TAG2  |       | TAG4  |       | TAG5  |       |
|---------|-------|-------|-------|-------|-------|-------|-------|-------|
|         | JAG   | VI    | JAG   | VI    | JAG   | VI    | JAG   | VI    |
| <START> | 28.11 | 16.31 | 21.6  | 11.23 | –     | –     | –     | –     |
| PUNCT   | 22.78 | 21.39 | 4.44  | 3.74  | 2.07  | 1.6   | 7.99  | 5.08  |
| NOUN    | 14.5  | 19.77 | 3.55  | 4.28  | 2.37  | 4.01  | 7.99  | 14.97 |
| ADV     | 14.5  | 21.12 | 4.44  | 2.94  | 16.27 | 20.32 | 13.61 | 17.91 |
| PRON    | 3.85  | 3.48  | 7.4   | 3.21  | 2.96  | 3.48  | 6.51  | 6.42  |
| ADJ     | 3.25  | 2.67  | 0.59  | 0.8   | 1.78  | 2.94  | 3.55  | 6.95  |
| ADP     | 2.96  | 2.67  | –     | 0.53  | 8.58  | 11.23 | 14.5  | 17.11 |
| VERB    | 2.66  | 1.34  | 20.12 | 37.97 | 35.5  | 35.83 | 19.53 | 12.3  |
| PROP    | 2.37  | 2.94  | –     | –     | –     | 0.53  | 0.59  | 2.14  |
| DET     | 2.37  | 1.6   | –     | –     | 2.37  | 4.55  | 10.06 | 6.95  |
| CCONJ   | 1.18  | 4.55  | 7.4   | 6.15  | 2.07  | –     | 1.18  | 1.87  |
| INTJ    | 0.89  | 0.53  | –     | –     | –     | –     | 0.3   | –     |
| AUX     | 0.3   | 0.8   | 15.68 | 11.23 | 18.34 | 8.29  | 4.44  | 2.94  |
| SCONJ   | 0.3   | 0.27  | 14.79 | 17.91 | 0.3   | 1.34  | 2.66  | 0.27  |
| NUM     | –     | 0.53  | –     | –     | 1.18  | 2.14  | 1.48  | 2.4   |
| PART    | –     | –     | –     | –     | 6.21  | 3.74  | 5.33  | 2.67  |
| <END>   | –     | –     | –     | –     | –     | –     | 0.3   | –     |

Table 4: Normalized POS-tag fivegram frequencies for *jag* and *vi* across TAG1 and TAG2 (before the pronoun), TAG4 and TAG5 (after the pronoun).

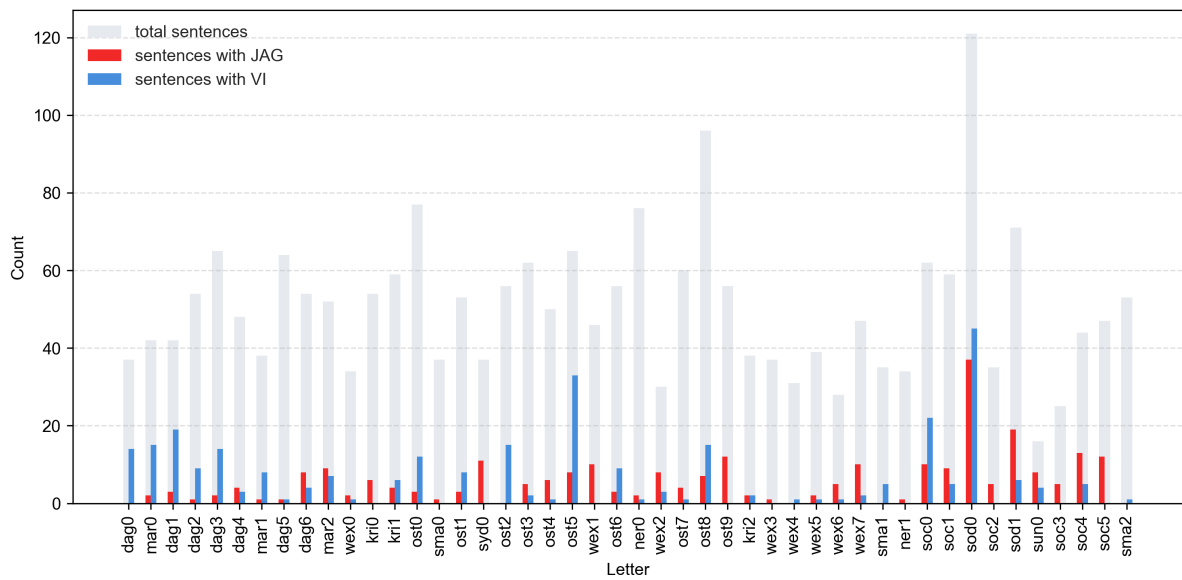


Figure 3: Count of *jag*- and *vi*-sentences as well as total sentences per letter, in chronological order.